

Student Performance Prediction

Final Capstone Project

AI/ML Final Assignment - Week 6

Domain: Artificial Intelligence | Batch: 6-Week AI Internship

Assignment Type: Final Project | Total Marks: 100

1. Introduction

This report presents the final capstone project of a 6-week AI internship, applying the complete Machine Learning workflow covered across the program: Problem Definition, Data Collection, Data Cleaning, Data Analysis, Model Development, Training, Prediction, Evaluation, and Conclusion. The project brings together Python, Pandas, NumPy, Matplotlib/Seaborn, and Scikit-learn skills developed in Weeks 2 through 5 into one complete, self-contained AI/ML solution.

2. Problem Statement

Project Title: Student Performance Prediction

Problem statement: Given a student's demographic details and academic habits (study hours, attendance, previous marks, assignment scores, sleep habits, and extracurricular involvement), predict whether the student will **Pass** or **Fail**.

Objective: Build and compare multiple classification models capable of accurately predicting student outcomes, and identify which student characteristics are most strongly associated with passing or failing.

Why AI/ML is suitable for this problem: The relationship between study habits and academic outcome is influenced by multiple interacting factors (attendance, prior performance, effort, rest) that are difficult to capture with simple fixed rules. Machine Learning models can automatically learn the relative importance of each factor directly from historical data, and can be easily retrained as new student data becomes available.

Expected outcome: A trained classification model that can flag at-risk students early, based on measurable, actionable factors like study hours and attendance.

Real-world application: Schools and universities could use a system like this as an early-warning tool, allowing teachers or academic advisors to identify and support struggling students before final exams, rather than only discovering problems after grades are already final.

3. Dataset

Source: Synthetically generated for this capstone project, modeling realistic relationships between student habits and outcomes (continuing the student-performance theme developed across Weeks 2-5 of this internship).

Size: 155 student records (150 after removing 5 duplicate rows during cleaning), 10 columns.

Important features:

- Gender - student's gender (Male/Female)
- Age - student's age in years
- Study_Hours - average daily study hours
- Attendance - class attendance percentage
- Previous_Marks - marks from a prior assessment
- Assignment_Score - score on assignments
- Sleep_Hours - average daily sleep hours
- Extracurricular - whether the student participates in extracurricular activities (Yes/No)
- Result (target) - Pass or Fail

4. Data Preprocessing

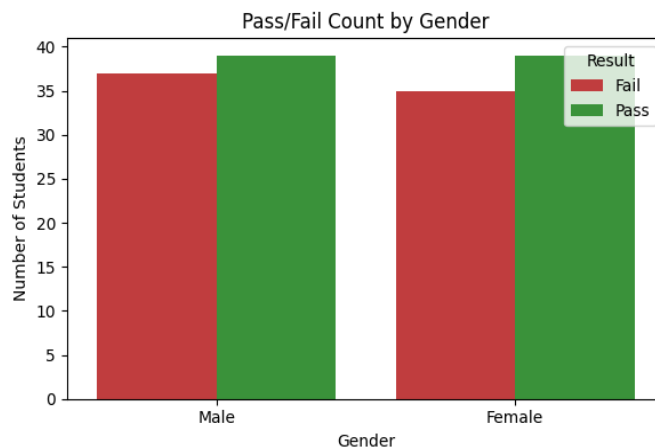
The dataset was loaded with Pandas and explored (first/last records, shape, column data types). A small number of missing values were found across five numeric columns (Study_Hours, Attendance, Previous_Marks, Assignment_Score, Sleep_Hours) and were filled using each column's mean, preserving all records rather than dropping rows. Five duplicate records were identified using ``duplicated()`` and removed, leaving 150 clean records.

Metric	Before Cleaning	After Cleaning
Total rows	155	150
Missing values	27	0
Duplicate rows	5	0

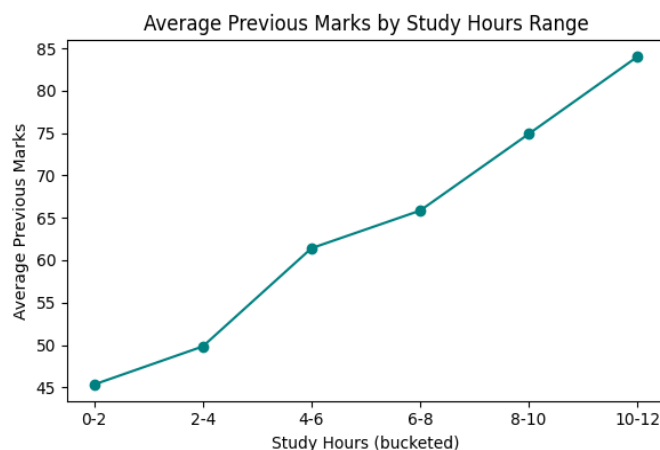
Two categorical columns (Gender, Extracurricular) were label-encoded into numeric form so they could be used by the ML models. Features (X) were separated from the target (y), and the target column, Result, was kept in its original Pass/Fail form for the classification models used (which handle string class labels natively in Scikit-learn).

5. Exploratory Data Analysis

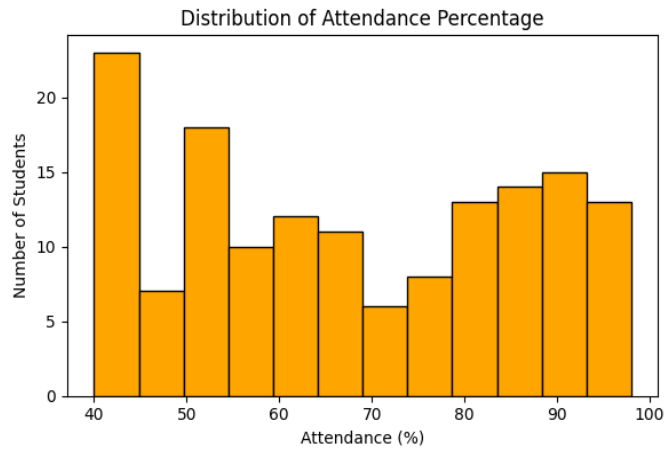
At least 5 visualizations were created to understand the dataset before modeling.



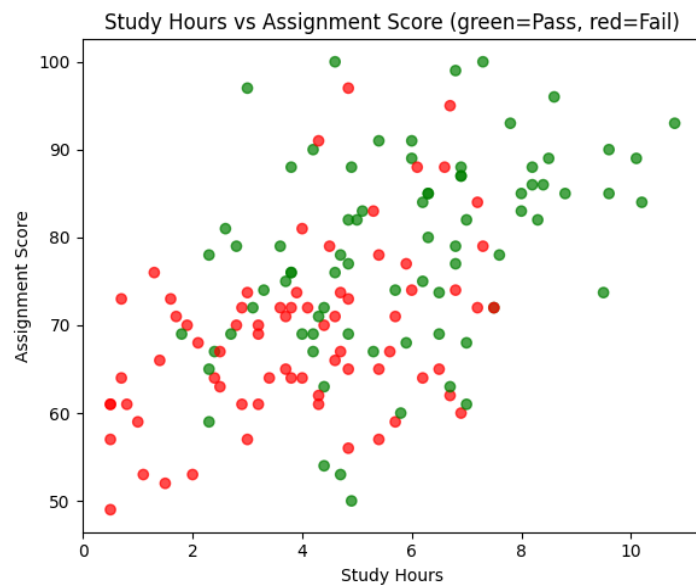
Bar Chart (Pass/Fail by Gender): Both genders show a broadly similar Pass/Fail split, suggesting gender is not a strongly determining factor here.



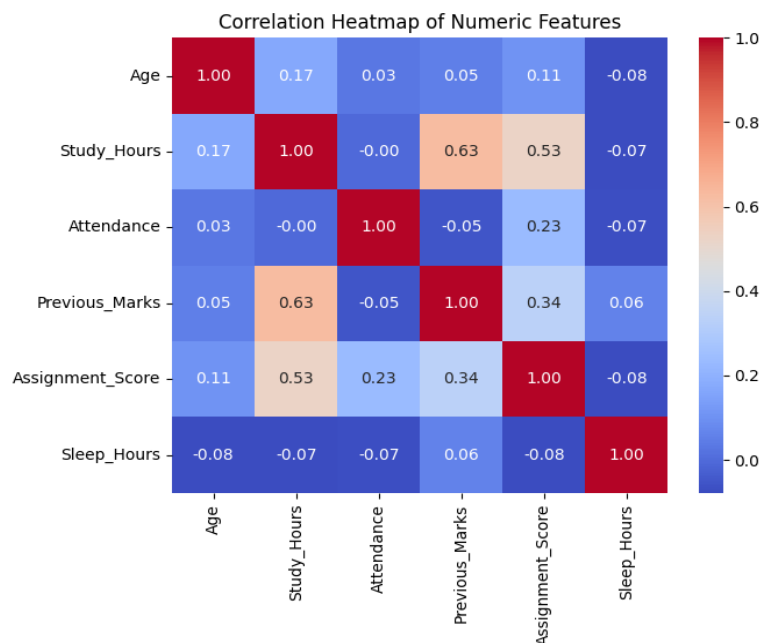
Line Chart (Study Hours vs Avg. Previous Marks): Average previous marks rise steadily as study hours increase, confirming a positive relationship between study effort and past performance.



Histogram (Attendance Distribution): Attendance is fairly spread out with a slight lean toward the higher end, indicating most students attend regularly but a meaningful minority have low attendance worth flagging.



Scatter Plot (Study Hours vs Assignment Score, colored by Result): Passing students (green) cluster toward higher study hours and assignment scores, while failing students (red) cluster toward the lower-left, visually confirming these features carry real predictive signal.



Correlation Heatmap: Study Hours, Attendance, Previous Marks, and Assignment Score are all positively correlated with each other. Age and Sleep Hours show much weaker correlation, suggesting less standalone predictive power.

6. Model Development

Data was split 80/20 into training (120 students) and testing (30 students) sets, stratified to preserve the Pass/Fail balance. Features were standardized for Logistic Regression using `StandardScaler`. Three classification models were selected and trained:

- **Logistic Regression** - a simple, fast, interpretable baseline for binary classification.
- **Decision Tree Classifier** (max depth 5) - captures non-linear, rule-based relationships and is easy to explain.
- **Random Forest Classifier** (100 trees, max depth 6) - an ensemble of many decision trees, typically more accurate and more robust to overfitting than a single tree.

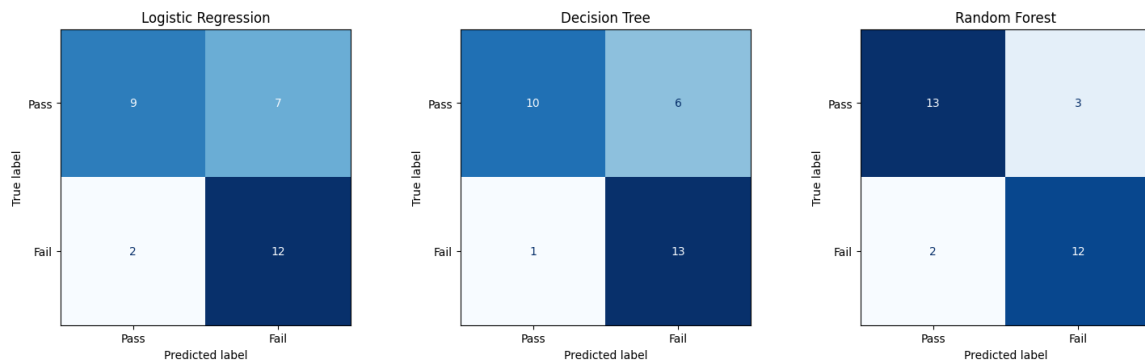
All three were chosen specifically so their performance could be directly compared, illustrating the trade-off between model simplicity/interpretability (Logistic Regression, Decision Tree) and predictive strength (Random Forest).

7. Training & Prediction

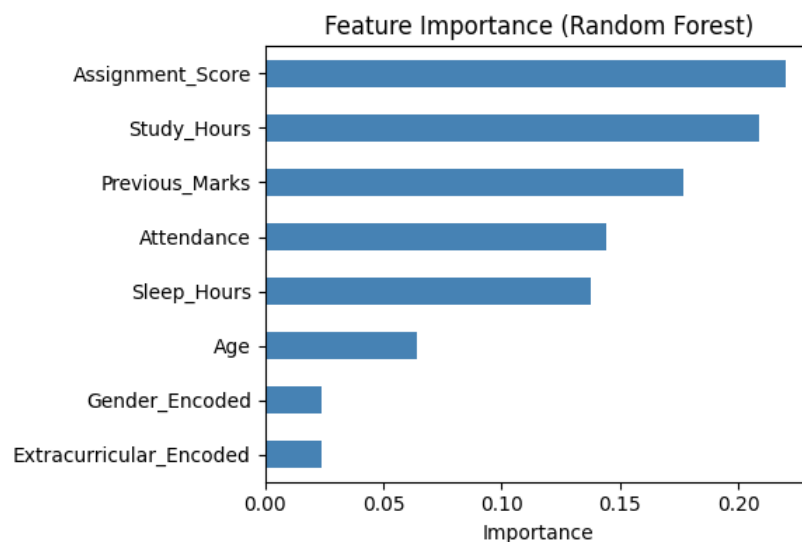
Each model was trained (`.fit()`) on the scaled/unscaled training data as appropriate, then used to generate predictions (`.predict()`) on the held-out test set of 30 students never seen during training. Predictions were compared against each student's actual Pass/Fail outcome to calculate performance metrics.

8. Model Evaluation

Model	Accuracy	Precision	Recall
Logistic Regression	70.00%	81.82%	56.25%
Decision Tree	76.67%	90.91%	62.50%
Random Forest	83.33%	86.67%	81.25%



Confusion matrices for all three models (test set, n=30).



Feature importance from the Random Forest model. Assignment_Score, Study_Hours, and Previous_Marks are the most influential predictors, consistent with the correlation heatmap and scatter plot findings from the EDA section.

9. Conclusion

This project successfully built and compared three classification models to predict student Pass/Fail outcomes from study habits and academic history. **Random Forest was the best-performing model**, achieving the highest accuracy (83.33%) with balanced precision (86.67%) and recall (81.25%), outperforming both Logistic Regression (70.00%) and the single Decision Tree (76.67%). This matches the general expectation that ensemble methods like Random Forest reduce the overfitting risk of a single tree while still capturing non-linear relationships that Logistic Regression's linear boundary cannot.

The feature importance analysis and exploratory data analysis both consistently point to Assignment Score, Study Hours, Previous Marks, and Attendance as the strongest predictors of student outcome — actionable factors that an educator could realistically monitor and intervene on.

Limitations: The dataset, while reasonably sized for a class project (150 students), is still small relative to real institutional data, so accuracy figures carry some statistical uncertainty (the test set is only 30 students). The Pass/Fail label was generated with intentional randomness, capping the maximum accuracy achievable by any model. Results reflect a single train-test split rather than cross-validation.

10. Future Scope

- Use a larger, real-world dataset from an actual school/university to validate the findings.
- Apply k-fold cross-validation for a more robust, less split-dependent accuracy estimate.
- Use GridSearchCV or RandomizedSearchCV to tune each model's hyperparameters for potentially higher accuracy.
- Add richer features, such as socioeconomic background, prior GPA trend over time, or teacher feedback.
- Deploy the trained model as a simple web dashboard so teachers could input a student's current stats and get an early Pass/Fail risk prediction.

Deliverables Included

- This PDF report (full project write-up: introduction through future scope)
- Jupyter Notebook (Final_Project.ipynb) - Tasks 2-5, fully executed with outputs, charts, and 3 trained models
- Datasets: student_performance_final.csv (raw) and student_performance_cleaned.csv (cleaned)